WIKTORIA SZITENHELM

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Based in Poland | Open to Remote
Portfolio | LinkedIn | GitHub

Software engineer with experience building financial systems, distributed applications, and backend infrastructure across enterprise and Web3 environments.
Worked at J.P Morgan on modernising financial systems within the derivatives domain and developed distributed cross-chain systems across EVM and Solana ecosystems.

Strong interest in Python, system design, data-intensive applications, and solving complex engineering problems in fast-paced environments.

EDUCATION

University of Aberdeen: 2020 - 2025
Master of Science in Computing Science with Spanish (Sustained studies) - First Class
Dissertation Project: "Solidity vs Rust in Smart Contract Development"
- Compared Solidity and Rust for smart contract development through a real e-voting protocol.

Undergraduate Teaching Assistant (Computing Science) 09/2021 – 05/2023
- Assisted students in **Python** programming and problem-solving exercises.
- Explained technical concepts clearly and adapted explanations to different levels of experience.
- Supported workshops and practical lab sessions in a collaborative academic environment.

WORK EXPERIENCE

J.P. Morgan | Glasgow, United Kingdom 06/2023 - 06/2024
Software Engineer
- Contributed to migrating a legacy financial **Sybase/PowerBuilder** system into a modern **microservice-oriented architecture** using **React** (frontend) and **Spring Boot** (backend) with **Java**.
- Designed and developed **backend and frontend** components using Spring Boot and React, implementing **RESTful APIs** for seamless integration between the two layers.
- Worked with **AWS cloud** infrastructure and Terraform-based infrastructure configuration.
- Designed database schemas and participated in data migration workflows to PostgreSQL.
- Developed and executed comprehensive **unit and acceptance tests.**
- Worked within an **Agile Scrum** team, participating in daily stand-ups, sprint planning, and code reviews.
- Wrote clear and concise technical **documentation for the code** to ensure knowledge transfer.

XSwap | Remote 05/2025 - 10/2025
Smart Contract Engineer
- Built a distributed cross-chain messaging and asset transfer system connecting **EVM and Solana** ecosystems using Chainlink CCIP.
- Implemented modular message validation, decoding, and execution flows with strong emphasis on reliability, safety, and failure handling.
- Developed simulation scripts and engineering tooling to improve testing and developer productivity.
- Delivered clean, well-documented code with Git-based workflows and JavaScript simulation scripts.

Freelance Software Engineer 06/2025 - present
- Developing backend systems and smart contract applications using Rust, Solidity and TypeScript.
- Building API integrations, testing pipelines, and developer tooling.
- Building automation scripts and agentic workflows using Python, APIs, and modern AI tooling.

Mental Wellbeing Initiative - Founder 02/2026 - present
- Building an AI-assisted platform focused on conversational systems and scalable backend architecture.
- Leading **MVP planning**, product direction, user research, and technical architecture.

RELEVANT PROJECTS

ZenAgent (ETHGlobal Cannes, 2026): Built solo a decentralized AI wellness companion. Integrated World ID, 0G Storage, and ENS. Full Next.js 14 frontend with exercises, mood tracking, AI manifestation quotes via Anthropic API, and habit tracker. Delivered a functional prototype during the hackathon: ZenAgent
GitHub repo: https://github.com/wszitenhelm/zenagent

34us (ETHWarsaw, 2025): Won Base bounty building a Solidity dApp for peer-to-peer tutor matching with on-chain coordination and wallet integration.

WellKey (ZKHack Berlin, 2025): Won ZKPassport Bounty designing a privacy-preserving application focused on anonymous form submission using zero-knowledge proofs.

Solidity vs Rust in Smart Contract Development (2025):
- Designed and implemented a **Solidity-based e-voting protocol**, then replicated in Rust for comparison.
- Implemented commit-reveal scheme based on hashing and digital signatures.
- Compared execution models, security tradeoffs, and performance characteristics for Solidity and Rust.
- Created deployment scripts and test frameworks for both ecosystems.
- Demonstrated deep understanding of distributed systems, cryptographic primitives, and on-chain architecture and documented findings in an academic paper.

GitHub repo: https://github.com/wszitenhelm/e-voting-solidity-vs-rust

Python Projects:

NLP Pipeline (2025): Built text-processing workflows using spaCy, NumPy, NLTK; implemented POS tagging, similarity metrics, perplexity, and logistic regression classification.

Hash Cracking (2024): Implemented brute force, dictionary attacks, salted hash cracking, and rainbow tables; benchmarked performance across methods.
GitHub repo: https://github.com/wszitenhelm/hash_cracking

•**Automated planner (2022)**: Implemented the core functions e.g. heuristic function, function capable of validating a plan given a domain and a problem and finally implement heuristic search A*.

•**Algorithmic Problem Solving Moving Home (2022):** Developed a program for 2D/3D truck packing with boxes of various sizes and positions for optimised unpacking via different human chains.

•**Leaky Stack, Expression Binary Tree, SpaceZ Game (2021)**

TECHNICAL SKILLS

Languages: OOP fundamentals, Python**,** Java, JavaScript, TypeScript, Solidity, Rust

Backend & APIs: Spring Boot, Node.js, RESTful APIs, microservices architecture, distributed systems

Frontend: React, TypeScript

Cloud & Infrastructure: AWS Cloud Infrastructure, Terraform, CI/CD, PostgreSQL

Engineering: testing, Git workflows, Agile, AI-assisted development

Bilingual: Polish (native), English (fluent), Spanish (intermediate)

Community & Communication: Technical documentation, Demo scripts, Workshops

Blockchain & Web3: Ethereum, Solana, Chainlink CCIP

Markets & Trading Interests:
- Strong personal interest in crypto market structure, liquidity, derivatives, and trading psychology.
- Self-studying market concepts including liquidations, fair value gaps (FVGs), momentum indicators, and price action analysis.